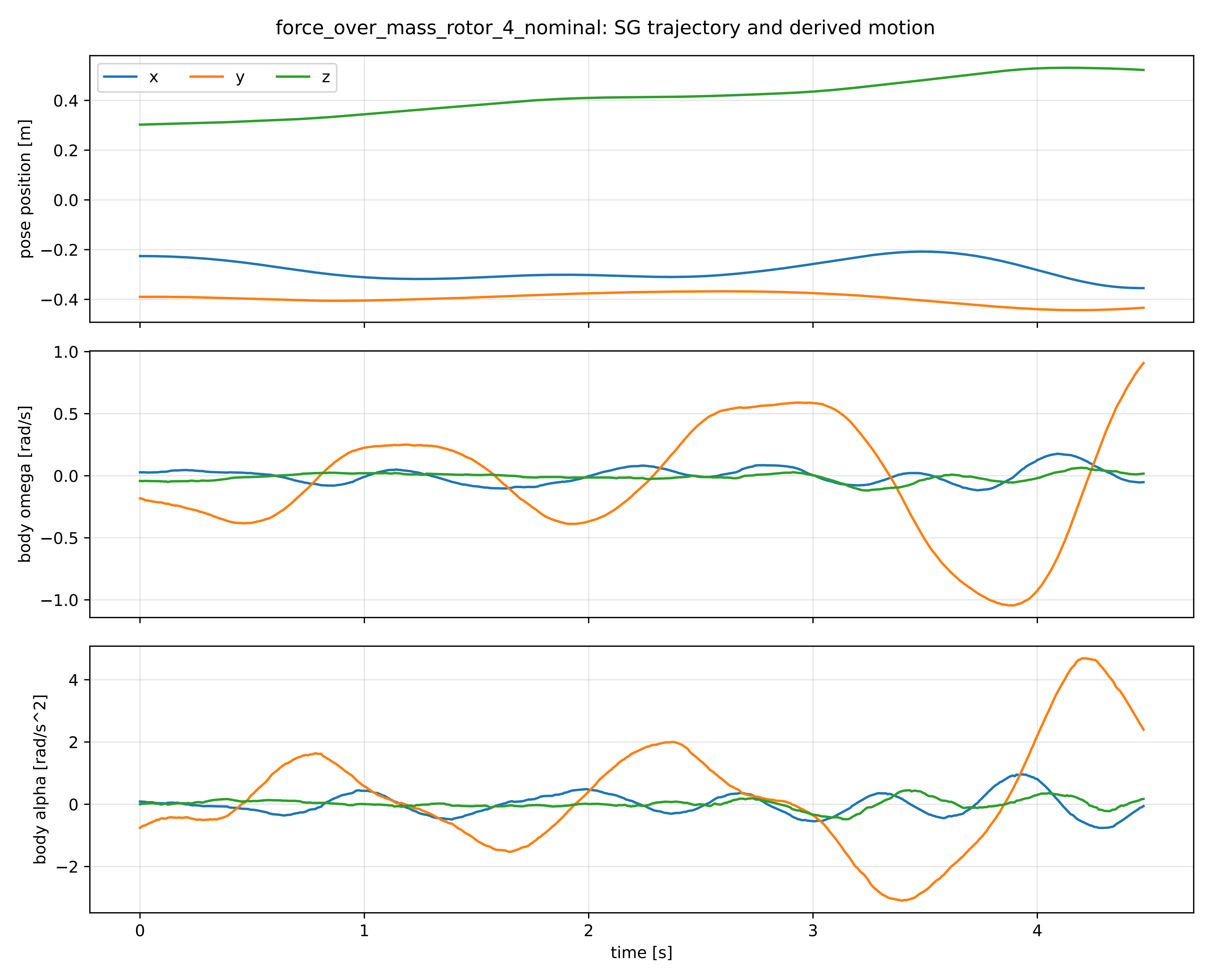
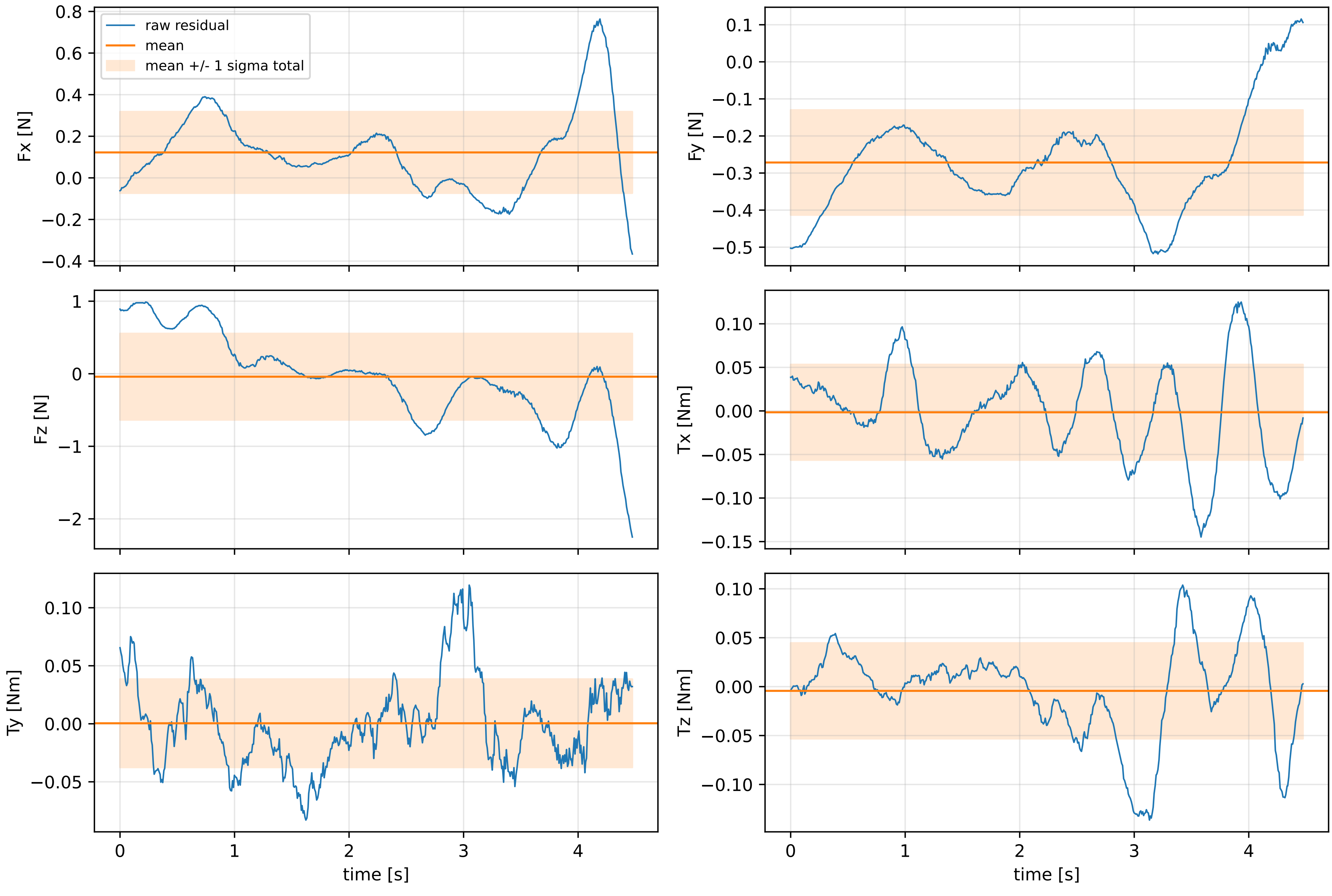
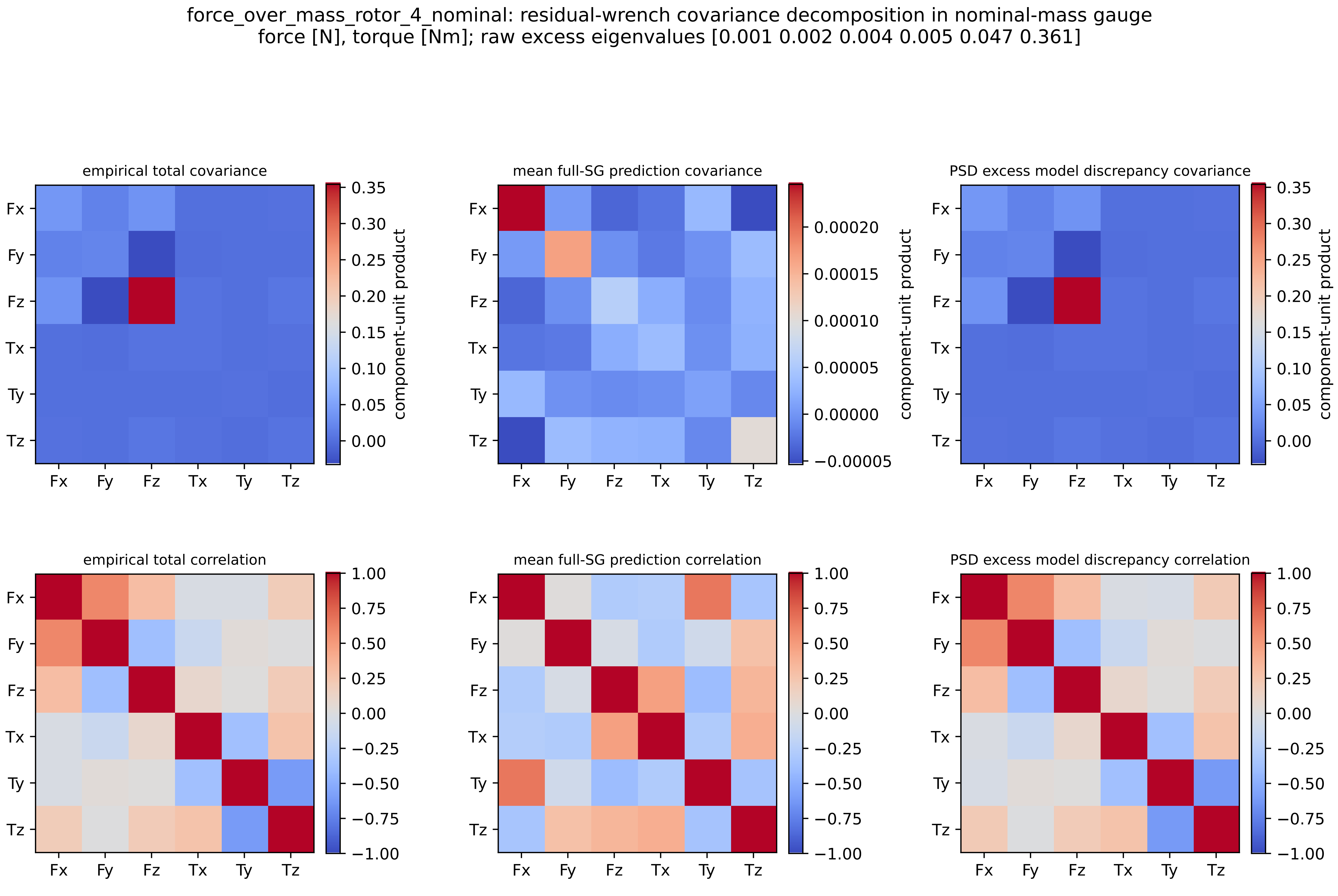
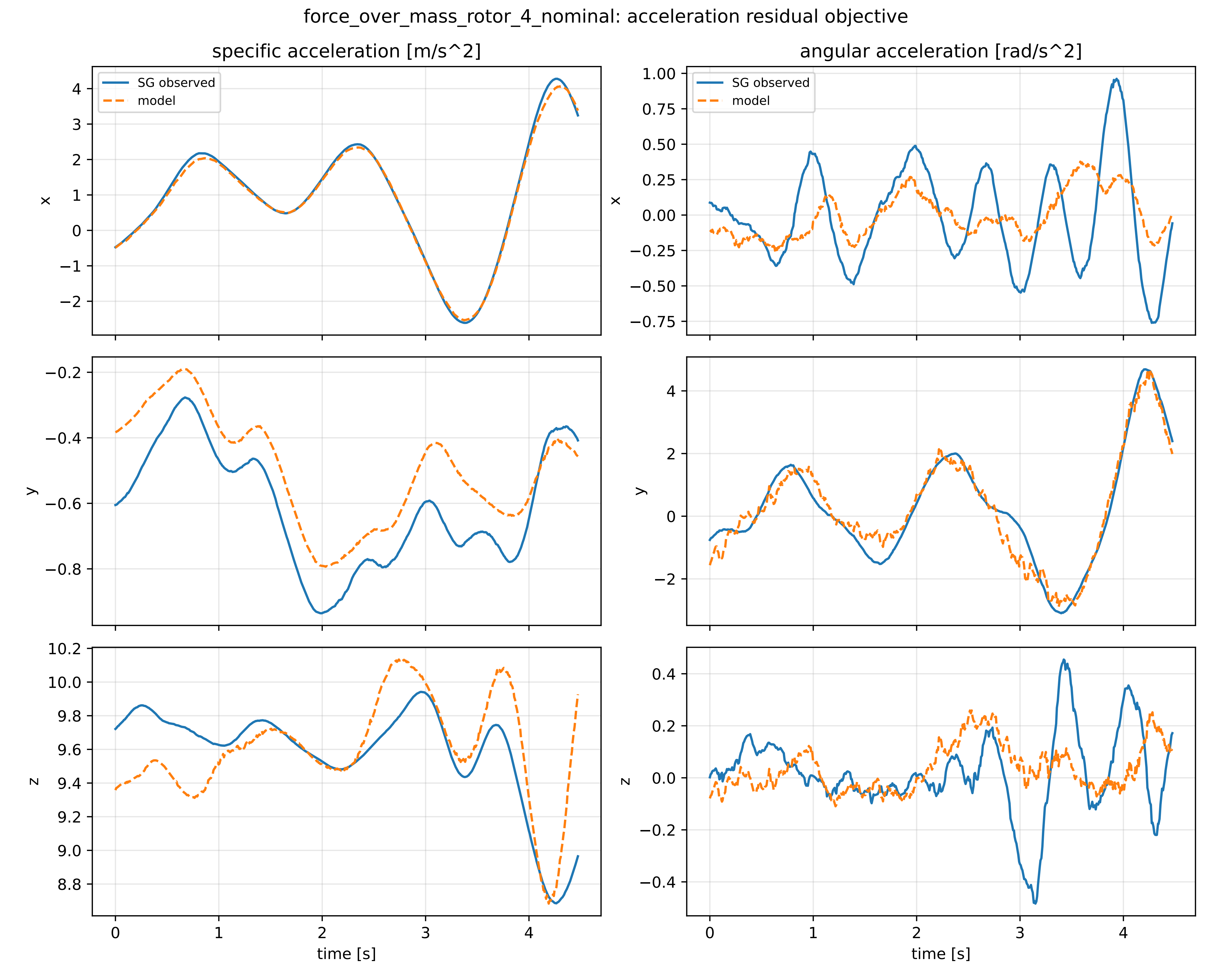

force_over_mass_rotor_4_nominal: SG trajectory and derived motion
x
y
z
pose position [m]
0.4
0.2
0.0
−0.2
−0.4
body omega [rad/s]
1.0
0.5
0.0
−0.5
−1.0
body alpha [rad/s^2]
4
2
0
−2
0
1
2
3
4
time [s]

force_over_mass_rotor_4_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg

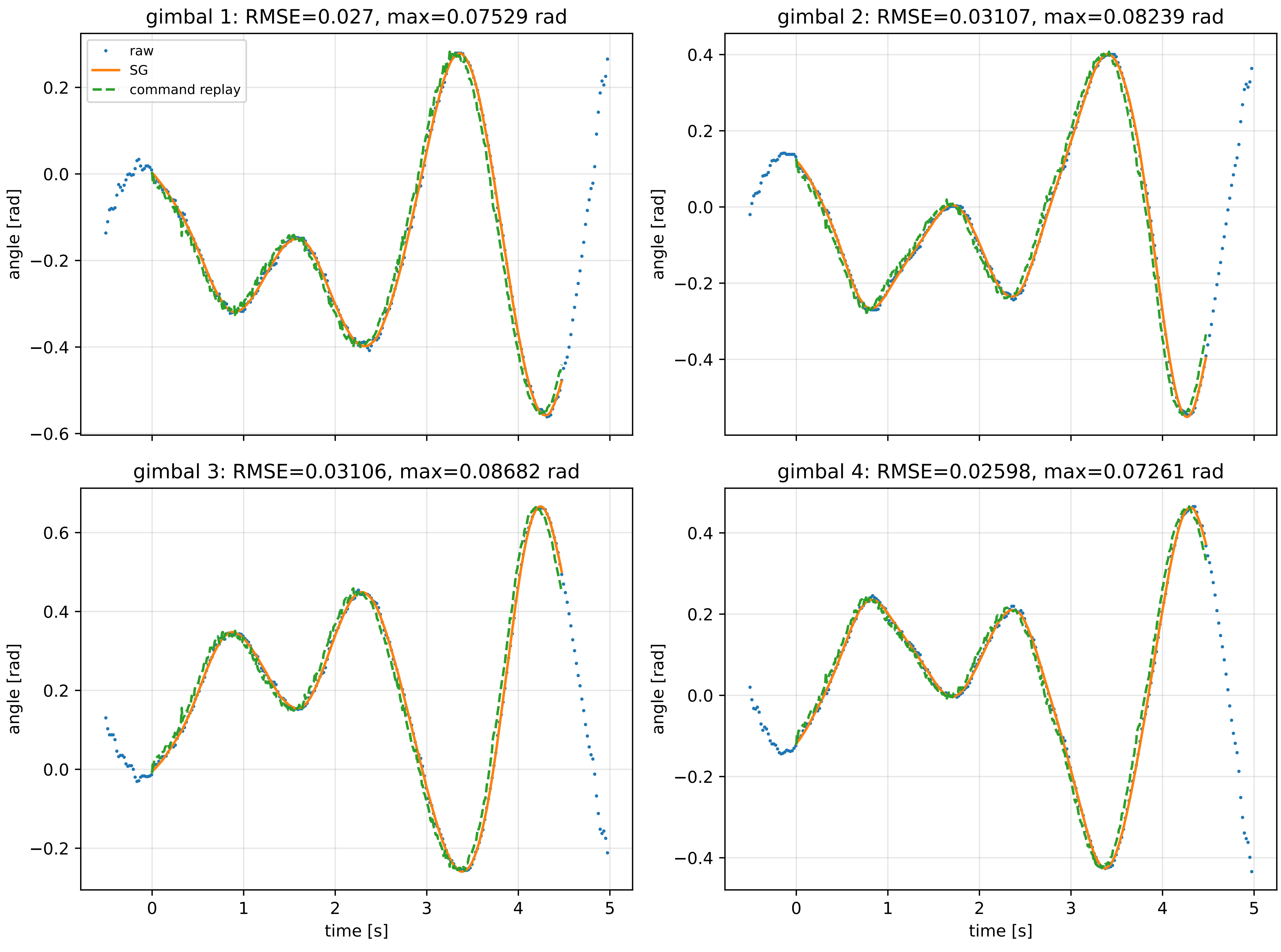


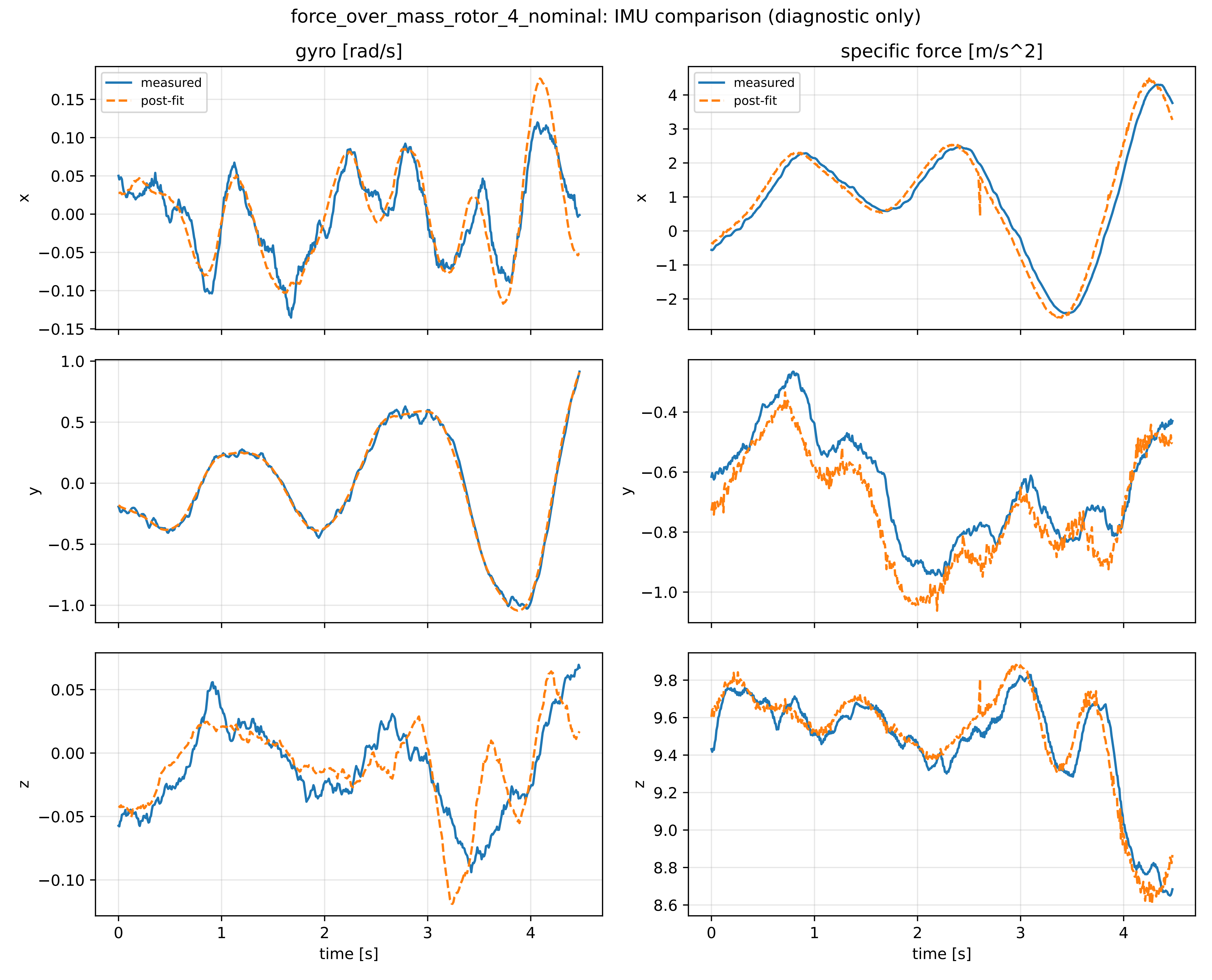
vector RMS about zero: force 0.7086 N, torque 0.082784 Nm; component std about mean: force [0.1957 0.1416 0.5952], torque [0.0548 0.038 0.0489]

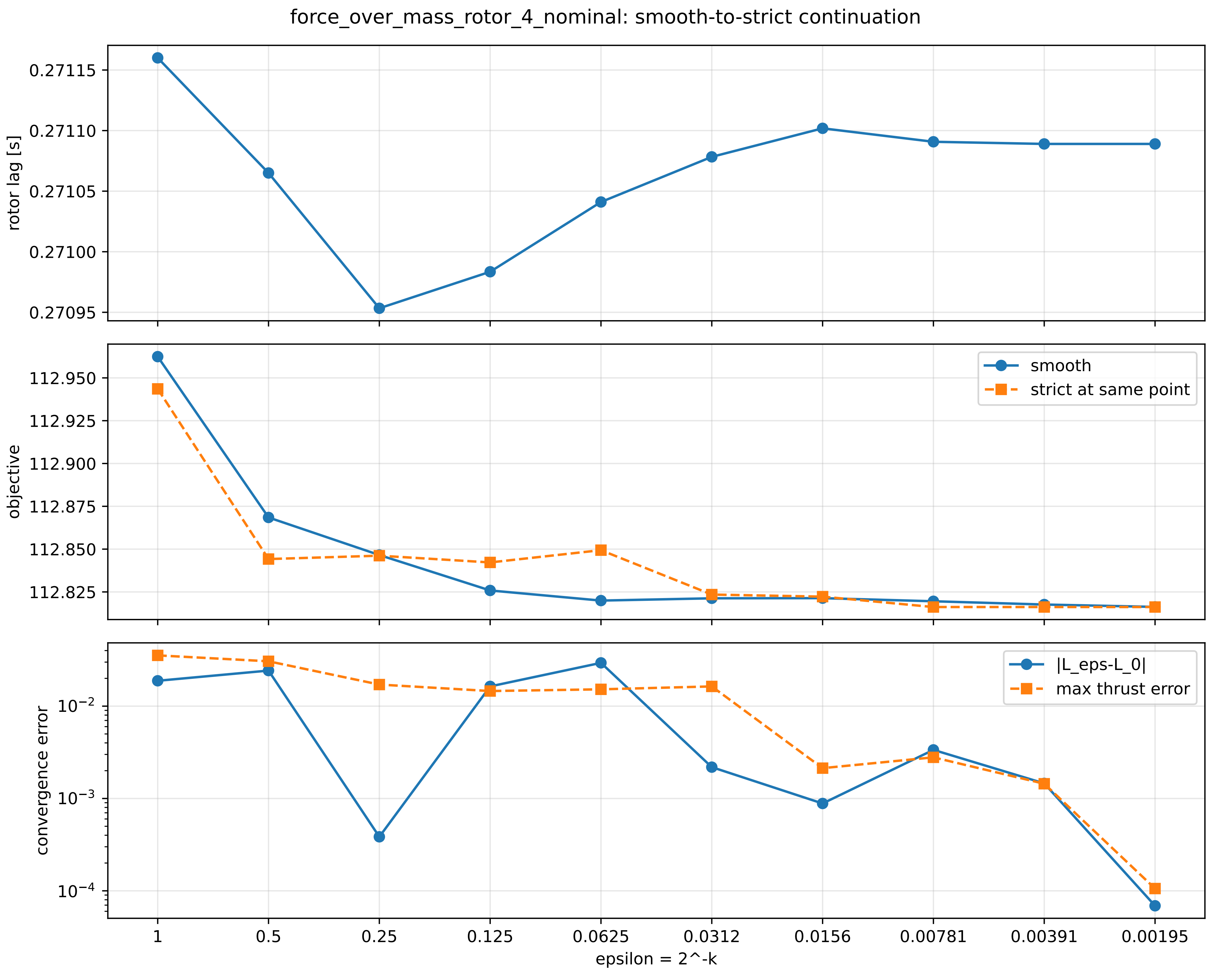

force_over_mass_rotor_4_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.004 0.005 0.047 0.361]
empirical total covariance
mean full-SG prediction covariance
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
Fx
Fy
Fz
Tx
Ty
Tz
component-unit product
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.00020
0.00015
0.00010
0.00005
0.00000
−0.00005
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


force_over_mass_rotor_4_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]

force_over_mass_rotor_4_nominal: gimbal measurement audit (objective=measured_sg)

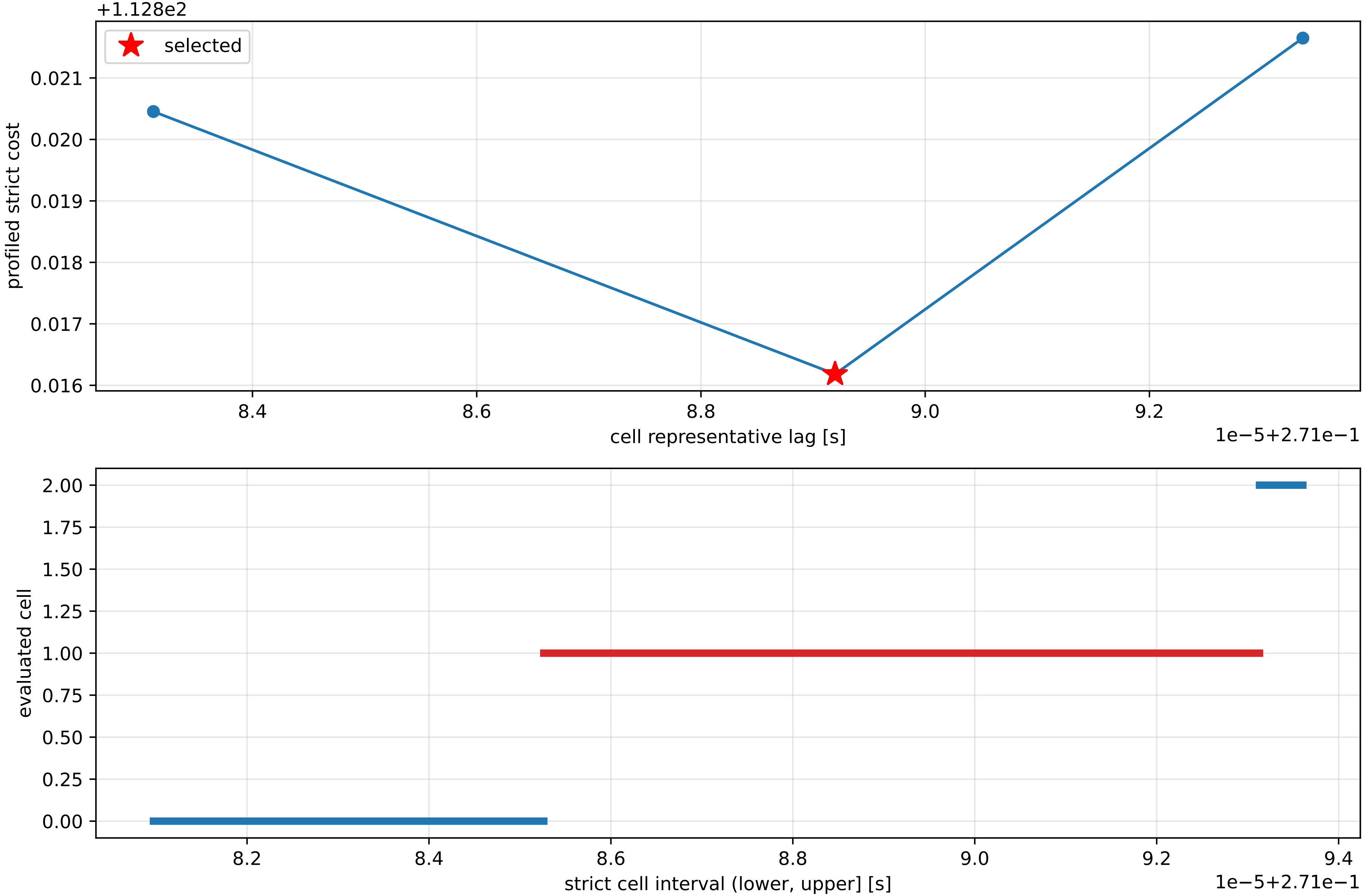



force_over_mass_rotor_4_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
measured
post-fit
x
y
z
x
y
z
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
1.0
0.5
0.0
−0.5
−1.0
0.05
0.00
−0.05
−0.10
4
3
2
1
0
−1
−2
−0.4
−0.6
−0.8
−1.0
9.8
9.6
9.4
9.2
9.0
8.8
8.6
0
1
2
3
4
time [s]
0
1
2
3
4
time [s]



force_over_mass_rotor_4_nominal: smooth-to-strict continuation
rotor lag [s]
0.27115
0.27110
0.27105
0.27100
0.27095
objective
112.950
112.925
112.900
112.875
112.850
112.825
smooth
strict at same point
convergence error
10^{-2}
10^{-3}
10^{-4}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

force_over_mass_rotor_4_nominal: exact strict-ZOH lag cells

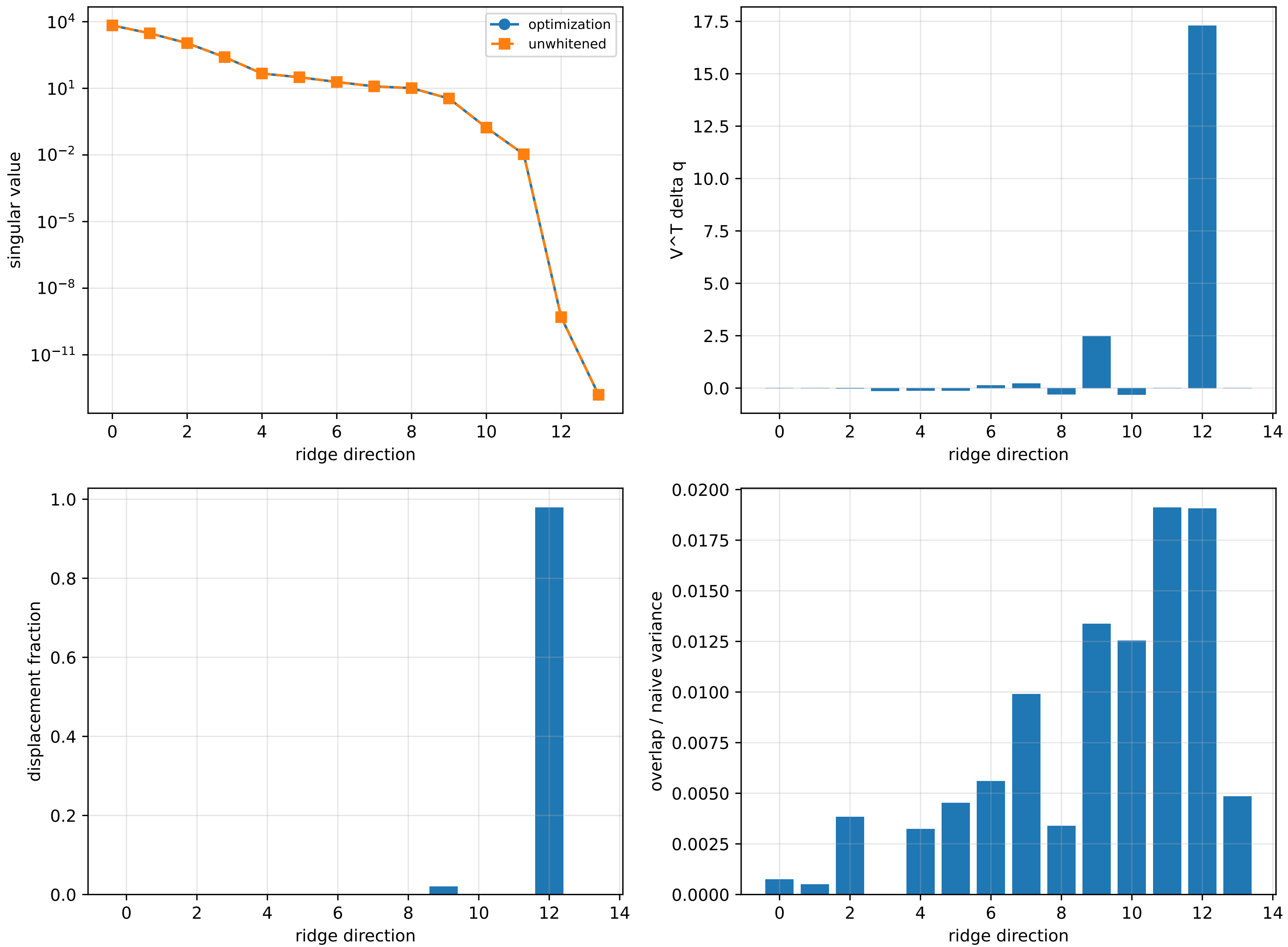


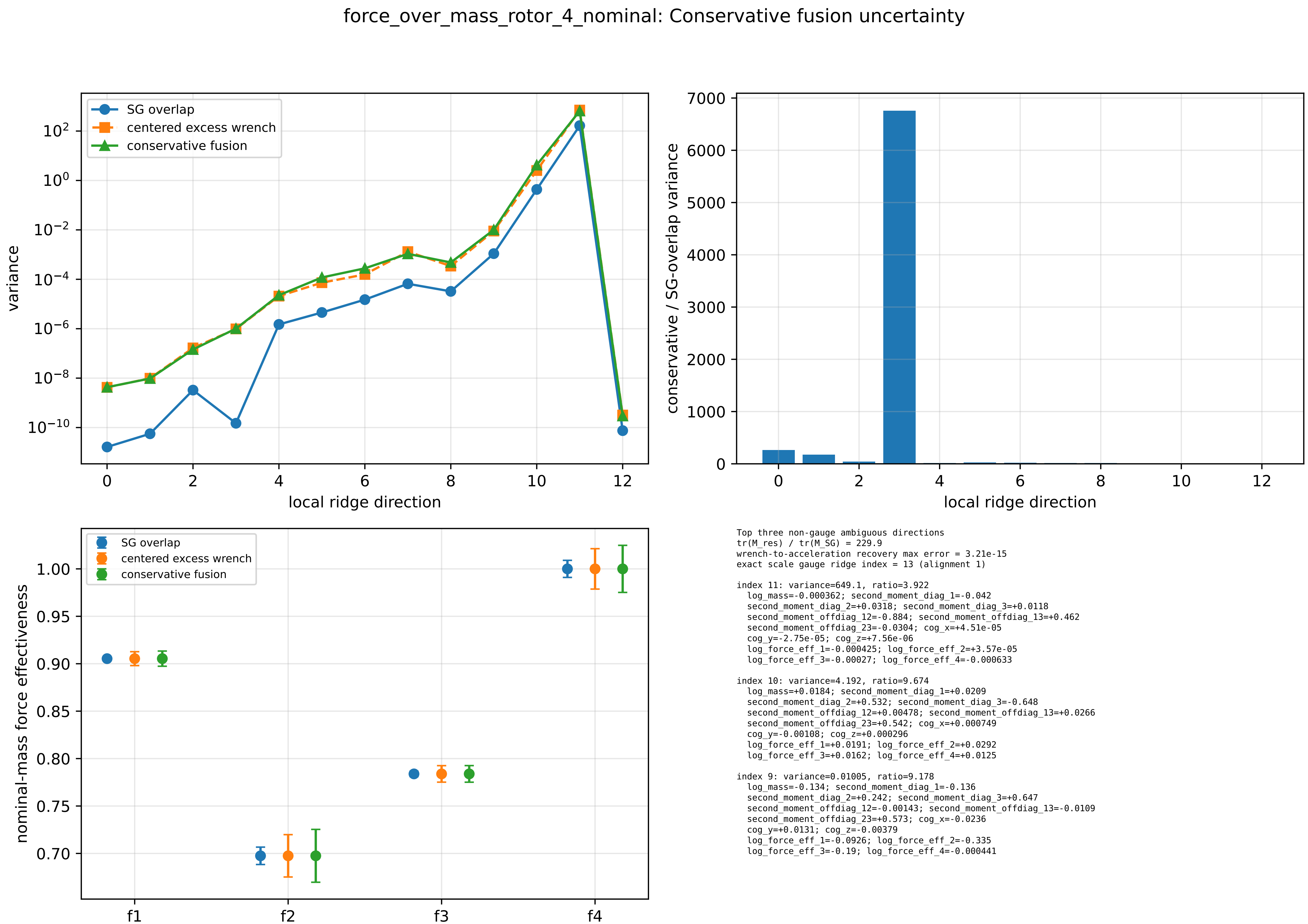
selected (0.271085262, 0.27109313] s; final smooth 0.271088991 s; data boundary=False

force_over_mass_rotor_4_nominal: parameter estimate

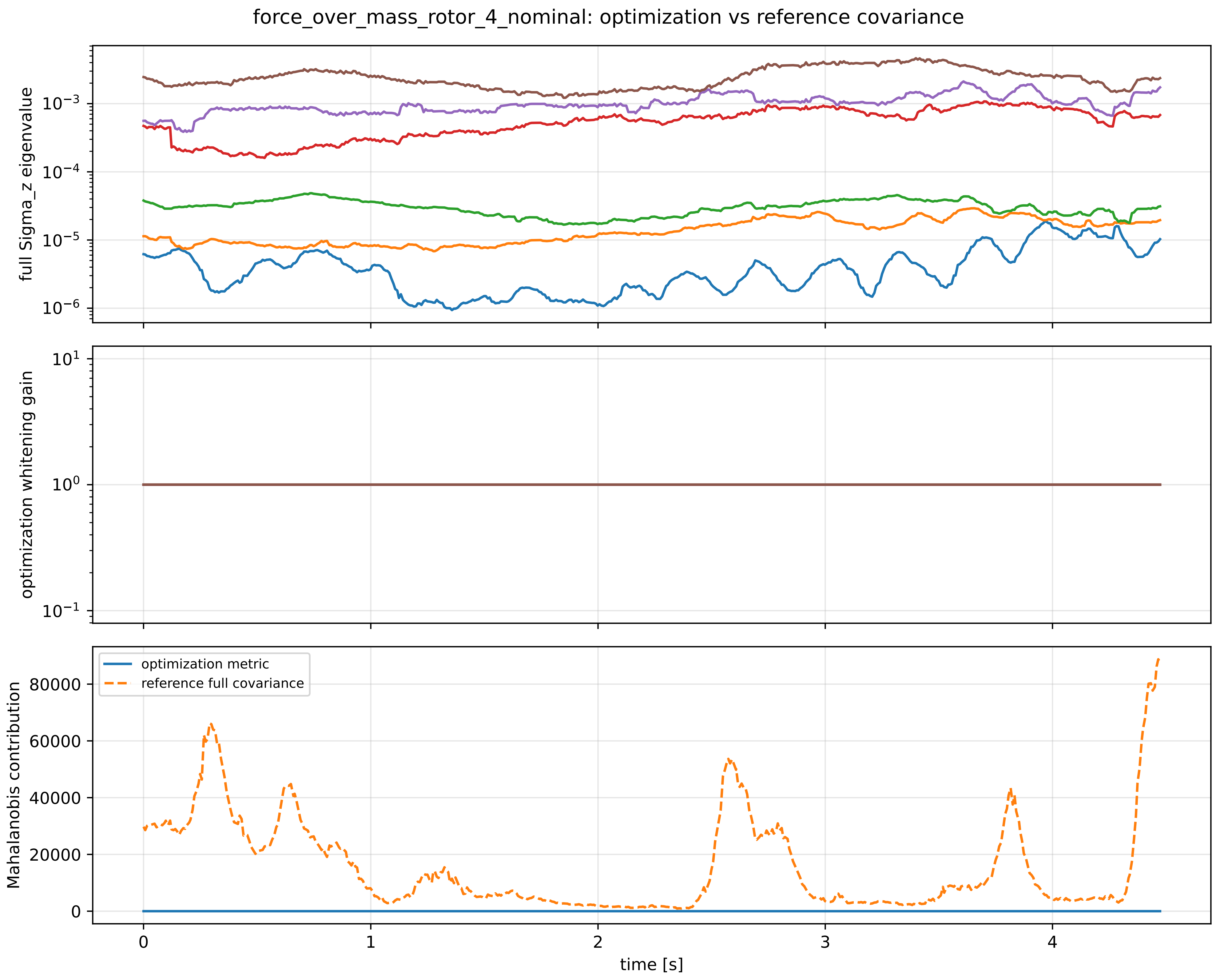
```
Case: force_over_mass_rotor_4_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 17.5875672
inertia [kg m^2]:
[[ 1.32636552  0.03730541  0.00309548]
 [ 0.03730541  0.62132385 -0.22612712]
 [ 0.00309548 -0.22612712  1.868162  ]]
CoG body [m]: [-0.03252514  0.01415968  0.0084814 ]
force effectiveness: [6.77120933 5.21652554 5.86244037 7.4789867 ]

scale-free J/m [m^2]:
[[ 0.07541495  0.00212112  0.000176  ]
 [ 0.00212112  0.03532745 -0.01285721]
 [ 0.000176   -0.01285721  0.1062206 ]]
scale-free f/m: [0.38499977 0.29660302 0.33332867 0.42524282]

rotor lag cell: (0.271085262, 0.27109313] s
representative: 0.271089196 s
```

force_over_mass_rotor_4_nominal: ridge spectrum (rank 12, nullity 2, ||Jv||=1.64e-13)

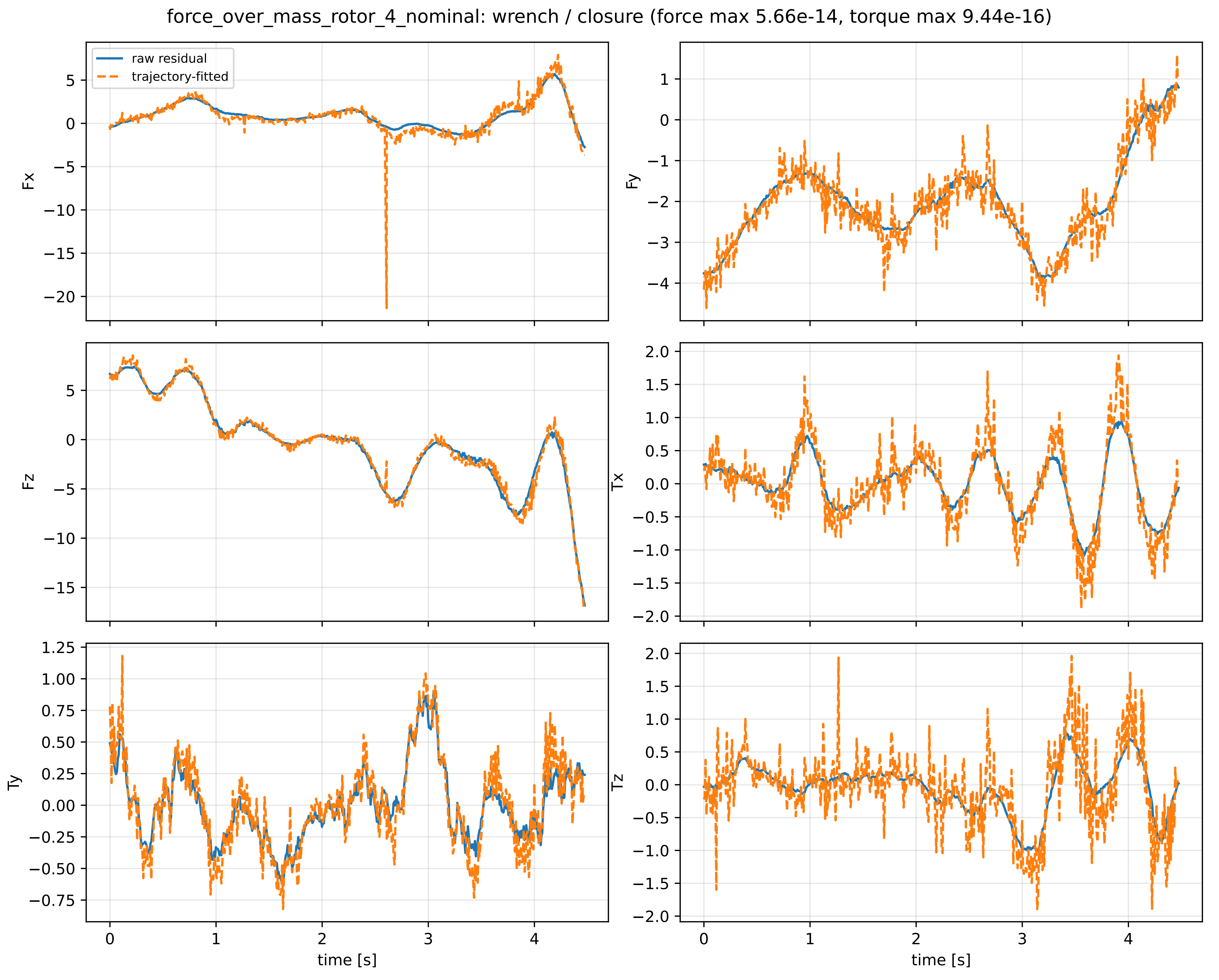




The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.



force_over_mass_rotor_4_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]


force_over_mass_rotor_4_nominal: wrench / closure (force max 5.66e-14, torque max 9.44e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

force_over_mass_rotor_4_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_force_over_mass_rotor_4_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass
SHA256: 678b0ccde06a50492269171f09bec596a62e0e49ae20754e771a2a2c2d4a9df2

data objective:  112.816182748
prior objective: 2.08106885157e-08
total objective: 112.816182769

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: force_over_mass_rotor_4_nominal (force_effectiveness_over_mass)
  components: rotor_4
  target: [0.42524282]
  std:    [1.e-05]
  final:  [0.42524282]
  error:  [-2.04013179e-09]
  standardized residual: [-0.00020401]
  factor objective: 2.08106885157e-08
```